Jay Fu

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Aug. 2026 - Jun. 2027**
Master of Science, Material Science and Engineering

Cornell University, College of Engineering, Ithaca, NY **Aug. 2023 - Jun. 2026**
Bachelor of Science, Material Science and Engineering
GPA: 3.5

Relevant Courses: Material Chemistry, Thermodynamics of Condensed Systems, Mechanical Properties of Materials, Polymer Mechanics, Kinetics, Diffusion, and Phase Transformations, Design of Soft Materials, Interfacial and Surface Phenomena in Materials Science, Physical Metallurgy and Applications, Introduction to Additive Manufacturing, Biomineralization: The Formation and Properties of Inorganic Biomaterials.

RELEVANT EXPERIENCE

Laboratory of Soft and Living Materials, Cornell University, *Graduate Researcher* **July. 2026 – Jun. 2027**

- Investigated acrylate-based chemistry to optimize polymerization kinetics and material performance.
- Designed and executed experiments to optimize polymer formulations and material performance.
- Characterized polymer materials using DMA, DSC, NMR, and GPC to establish structure-property relationships.
- Developed MATLAB tools for polymer characterization and experimental data analysis.

Laboratory of Soft and Living Materials, Cornell University, *Undergraduate Researcher* **Mar. 2025 – Jun. 2026**

- Standardized SOPs for PDMS film fabrication and characterization.
- Analyzed material structure-property relationships using CHESS synchrotron characterization techniques.
- Engineered and optimized polymer film formulations and manufacturing processes.

Cornell University Solar Boat Hull Team, Cornell University, *Subteam Member and Lead* **Oct. 2023 – Dec. 2025**

- Evaluated polymer-composite fabrication methods through prototype manufacturing and testing of hull structures.
- Collaborated with multidisciplinary engineering teams to coordinate design, fabrication, and testing activities.
- Presented technical updates, testing results, and project milestones to team leadership on a biweekly basis.

ADDITIONAL EXPERIENCE

Cosmos Co, Cornell University, *Marketing Lead* **Jan. 2025 - Jun. 2026**

- Developed and optimized small-scale polymer film formulations and manufacturing processes.
- Conducted technical and commercial analyses for sustainable food-packaging applications.
- Received Cornell MSE Design Lab Award for project feasibility and technical presentation.

Office of Inclusive Excellence, Cornell University, *AEW Facilitator* **Aug. 2025 - Dec. 2025**

- Organized activities for students to increase understanding of mathematical concepts.
- Facilitated small groups of students to work together and complete tasks through discussion.
- Worked with a co-facilitator to create and curate problems and concepts to demonstrate to the students.

SPECIALIZED SKILLS

Materials & Characterization: DMA, DSC, NMR, GPC, Polymer Processing, Polymer Synthesis
Programming: MATLAB, Python
CAD: Onshape(CAD), SolidWorks (CAD), Fusion 360 (CAD), MeshLab, Meshmixer, 3D Slicer, Dragonfly
Communication: Overleaf (LaTeX), Microsoft Workplace systems, Google Workplace systems
Fabrication Skills: Operation of lathes and mills (Red Apron machinist at Cornell), experience with 3D printing, laser cutting, drills and sawing
Foreign Languages: Mandarin (fluent); Latin (intermediate)